



Lab No 1

Ubuntu Server Installation

Submitted To: Engr.Waqas Saleem

Submitted By: Laiba Tariq

Reg No: 2024-BSE-85

Course Code: SE-665

Section: B

GitHub Username: LaibaTariq085

Task 1

Create and Configure Your GitHub Account

Description:

Created GitHub profile with username **LaibaTariq085** and updated the display name to **Laiba Tariq** in Public Profile settings.

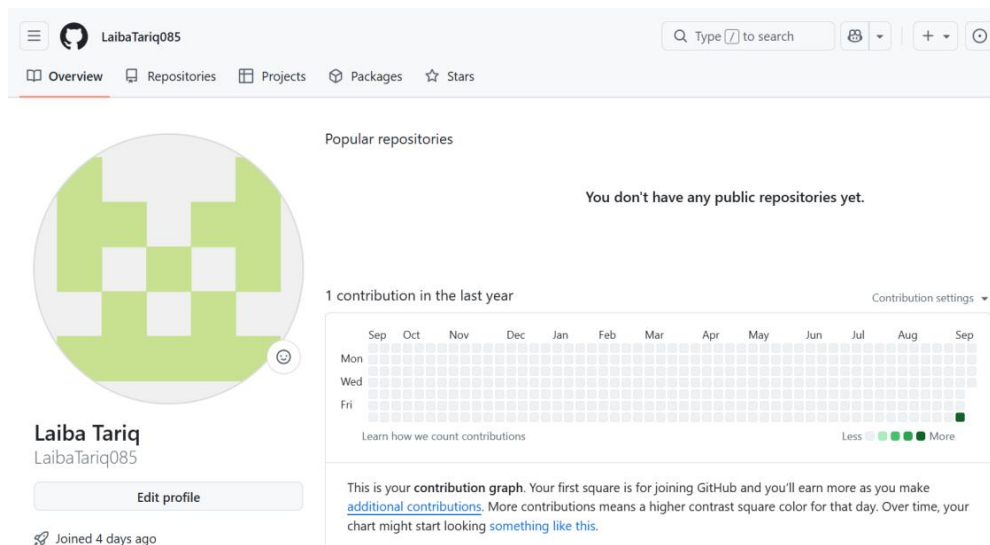


Figure 1.1: Public Profile (github_profile.png)

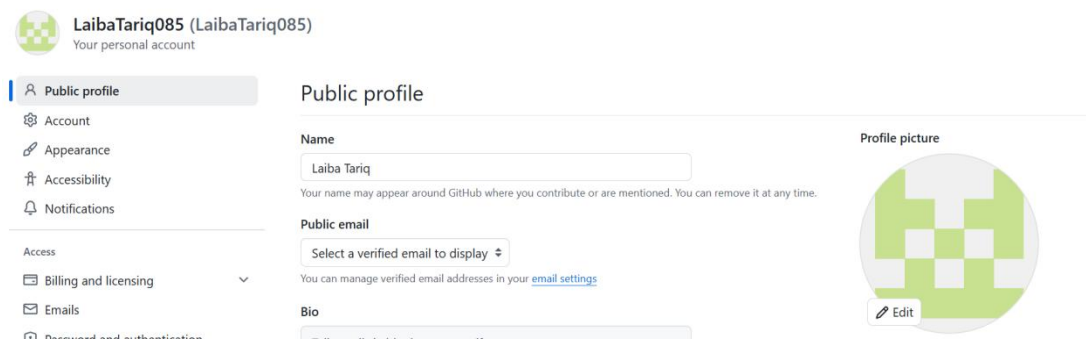


Figure 1.2: Profile Settings Name (github_actual_name.png)

Task 2

Register in the Student Marks Portal

Description:

Authenticated via GitHub on the Student Marks Portal and enrolled in Cloud Computing (SE-665), Section B using Registration No **2024-BSE-85**.

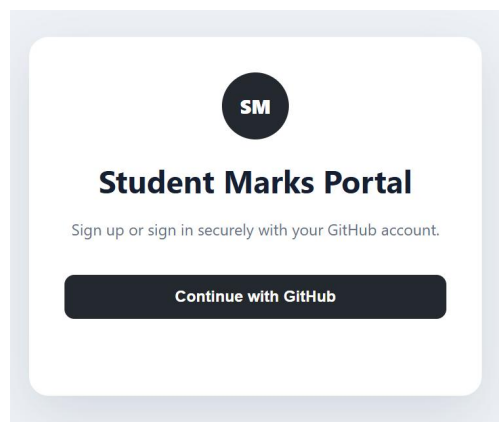


Figure 2.1: Portal Login Page (portal_login_page.png)



Figure 2.2: Authenticated Profile (portal_github_profile.png)

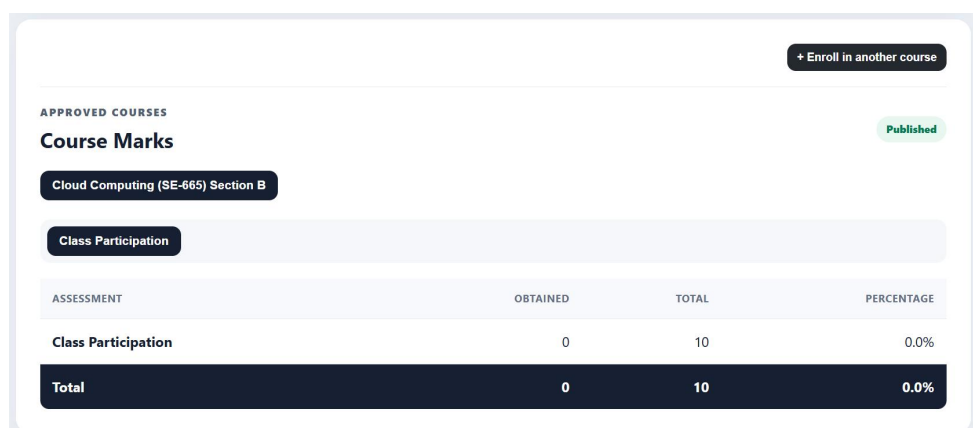


Figure 2.3: Course Enrollment Submitted (portal_enrollment_submitted.png)

Task 3

Download VMware Workstation Pro and Ubuntu Server

Description:

Installed VMware Workstation Pro on Windows host environment, verified download of the latest Ubuntu Server LTS ISO image, and confirmed sufficient disk storage capacity (minimum 20 GB) for virtual machine provisioning.

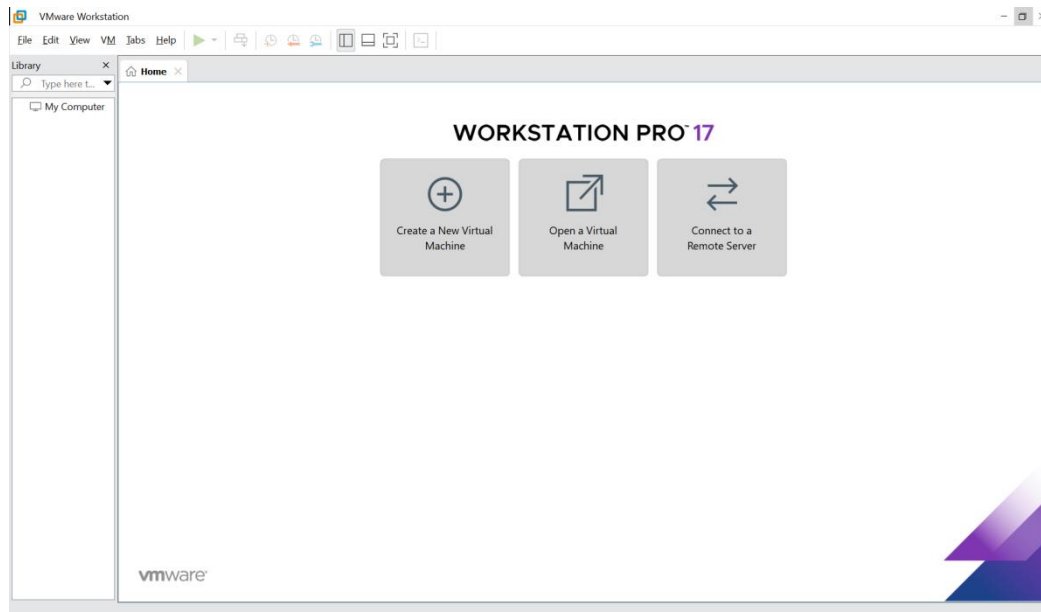


Figure 3.1: VMware Workstation Pro Interface (vmware_workstation.png)

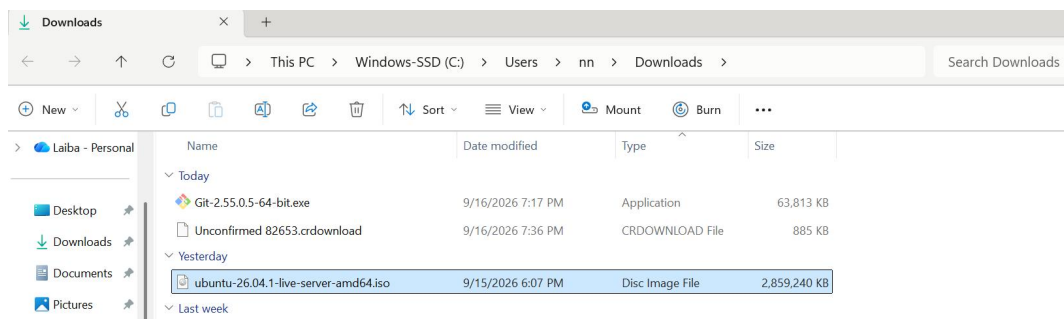


Figure 3.2: Downloaded Ubuntu Server ISO (ubuntu_server_iso.png)

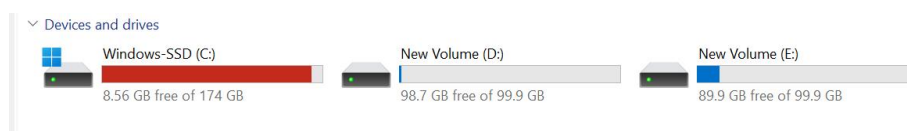


Figure 3.3: Available Free Disk Storage (available_storage.png)

Task 4

Create the Ubuntu Server Virtual Machine

Description:

Configured a new Virtual Machine in VMware Workstation Pro using the Typical wizard configuration, attached the downloaded Ubuntu Server ISO image, allocated 20 GB of virtual disk storage, and initiated the VM boot process.



Figure 4.1: New Virtual Machine Wizard (vm_creation_wizard.png)



Figure 4.2: Configuration Type (vm_typical_configuration.png)

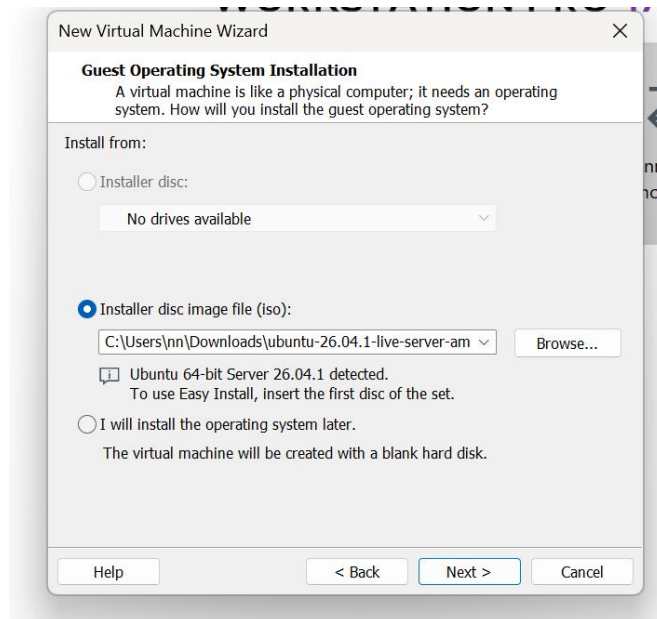


Figure 4.3: Selected ISO Path (vm_ubuntu_iso_selected.png)

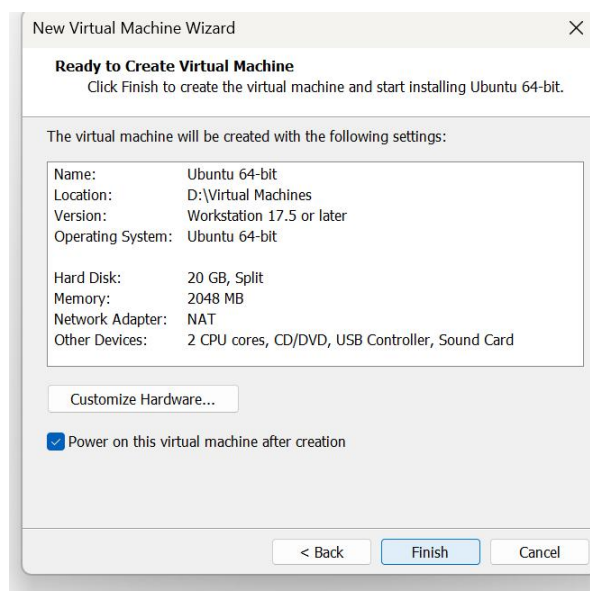


Figure 4.4: Configuration Summary (vm_configuration_summary.png)

```
[ OK ] Started dmesg.service - Save initial kernel messages after boot.
Starting dracut-shutdown.service - Restore /run/initramfs on shutdown...
Starting e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots...
[ OK ] Reached target getty.target - Login Prompts.
Starting networkd-dispatcher.service - Dispatcher daemon for systemd-networkd...
Starting nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot...
Starting polkit.service - Authorization Manager...
Starting rsyslog.service - System Logging Service...
[ OK ] Reached target getty-pre.target - Preparation for Logins.
Starting snapd.seeded.service - Wait until snapd is fully seeded...
Starting snapd.service - Snap Daemon...
Starting ssh.service - OpenBSD Secure Shell server...
Starting sysstat.service - Resets System Activity Logs...
Starting systemd-logind.service - User Login Management...
Starting udisks2.service - Disk Manager...
[ OK ] Finished blk-availability.service - Availability of block devices.
[ OK ] Finished dracut-shutdown.service - Restore /run/initramfs on shutdown.
[ OK ] Stopped ssh.service - OpenBSD Secure Shell server.
Starting ssh.service - OpenBSD Secure Shell server...
[ OK ] Finished nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot.
[ OK ] Reached target remote-fs-pre.target - Preparation for Remote File Systems.
[ OK ] Reached target remote-fs.target - Remote File Systems.
Starting apport.service - automatic crash report generation...
[ OK ] Started cron.service - Regular background program processing daemon.
Starting multipathd-queuing.service - Enable queuing for multipath maps...
Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Finished multipathd-queuing.service - Enable queuing for multipath maps.
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
[ OK ] Finished cloud-config.service - Cloud-init: Config Stage.
[ OK ] Started systemd-logind.service - User Login Management.
[ OK ] Started unattended-upgrades.service - Unattended Upgrades Shutdown.
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
[ OK ] Finished sysstat.service - Resets System Activity Logs.
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrgb.service - Set console scheme...
[ OK ] Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
[ OK ] Finished e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots.
[ OK ] Finished setvtrgb.service - Set console scheme.
[ OK ] Started rsyslog.service - System Logging Service.
[ OK ] Started chrony.service - chrony, an NTP client/server.
[ OK ] Started dbus.service - D-Bus System Message Bus.
[ OK ] Started polkit.service - Authorization Manager.
Starting ModemManager.service - Modem Manager...
[ OK ] Started udisks2.service - Disk Manager.
[ OK ] Started ModemManager.service - Modem Manager.
[ OK ] Finished apport.service - automatic crash report generation.
[ OK ] Started networkd-dispatcher.service - Dispatcher daemon for systemd-networkd.
-
```

Figure 4.5: Installer Boot Screen (ubuntu_installer_boot.png)

Task 5

Choose the Installation Language

Description:

Selected English as the primary installation language in the Ubuntu Server TUI wizard to proceed with the OS deployment.

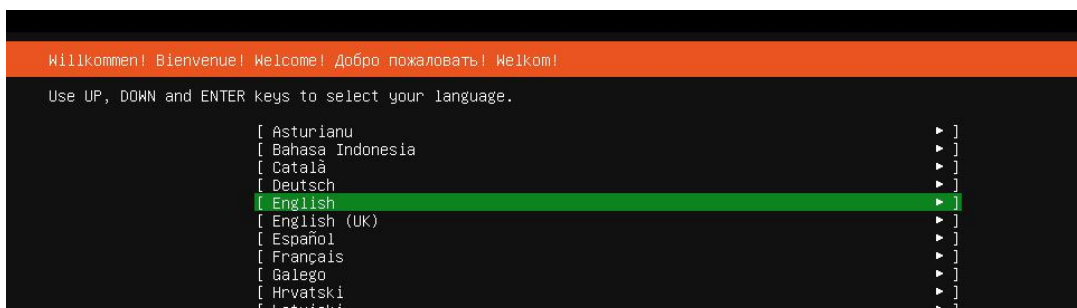


Figure 5.1: Selected Language (ubuntu_language.png)

Task 6

Configure the Keyboard Layout

Description:

Configured the primary keyboard layout and matching key variant settings for input compatibility during system operation.

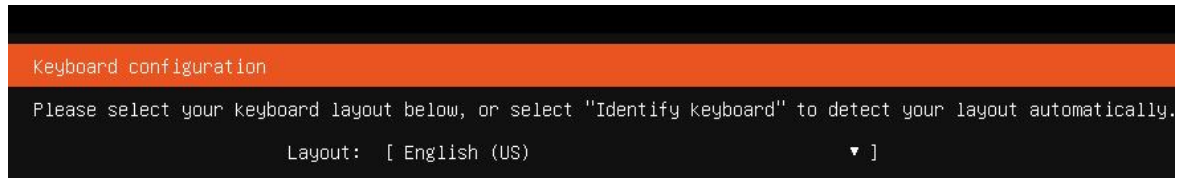


Figure 6.1: Selected Keyboard Layout (ubuntu_keyboard_layout.png)



Figure 6.2: Selected Keyboard Variant (ubuntu_keyboard_variant.png)

Task 7

Choose the Ubuntu Server Installation Type

Description:

Selected the standard base Ubuntu Server installation profile without additional MAAS or enterprise automated provisioning tools.

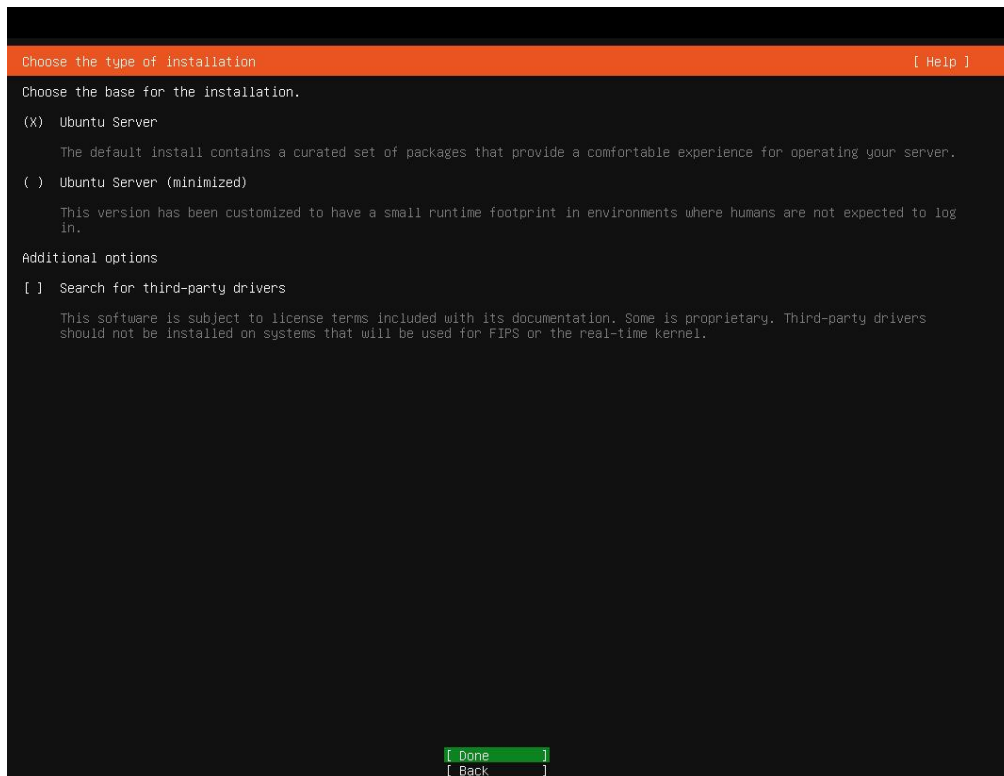


Figure 7.1: Selected Installation Type (ubuntu_installation_type.png)

Task 8

Configure the Network

Description:

Verified network interface detection by the installer and confirmed dynamic IPv4 allocation via DHCP for network connectivity.

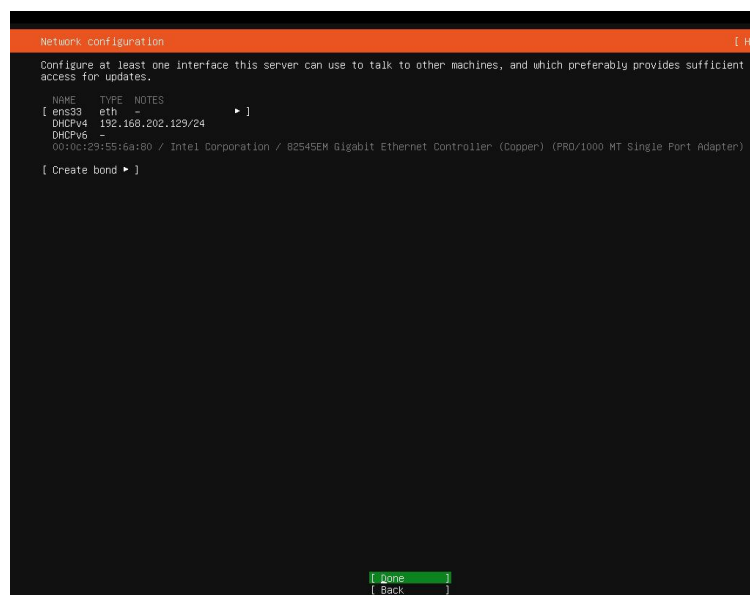


Figure 8.1: Detected Network Interface (ubuntu_network_interface.png)

```
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient
access for updates.

NAME      TYPE  NOTES
[ ens33   eth  -      ▶ ]
DHCPv4    192.168.202.129/24
DHCPv6    -
00:0c:29:55:6a:80 / Intel Corporation / 82545EM Gigabit Ethernet Controller (Copper) (PRO/1000 MT Single Port Adapter)

[ Create bond ▶ ]
```

Figure 8.2: Assigned Network Address (ubuntu_installer_network.png)

Task 9

Configure Storage

Description:

Selected the guided storage layout option to configure the VMware virtual disk for automated partition mapping and reviewed the resulting storage summary.

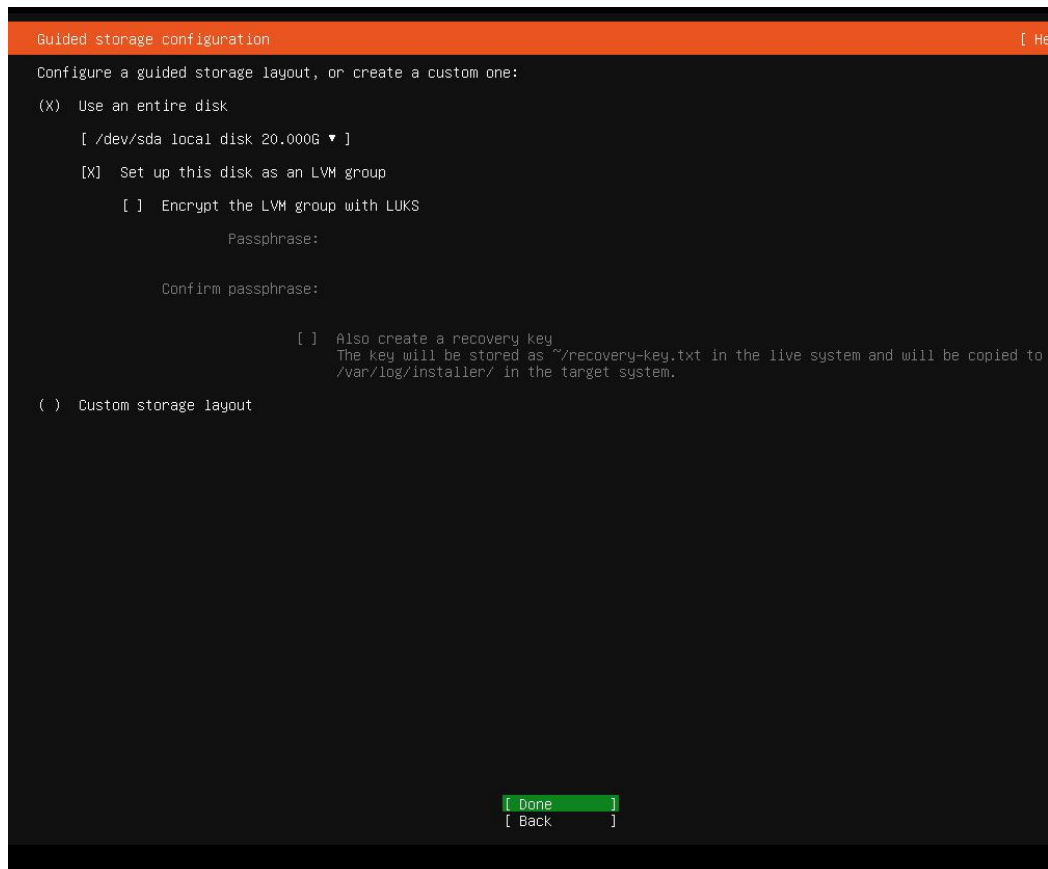


Figure 9.1: Guided Storage Option (ubuntu_guided_storage.png)



Figure 9.2: Storage Configuration Summary (ubuntu_storage_configuration.png)

Task 10

Select the Installation Disk

Description:

Identified and selected the designated VMware virtual drive, confirming its total capacity and volume details prior to formatting and system deployment.

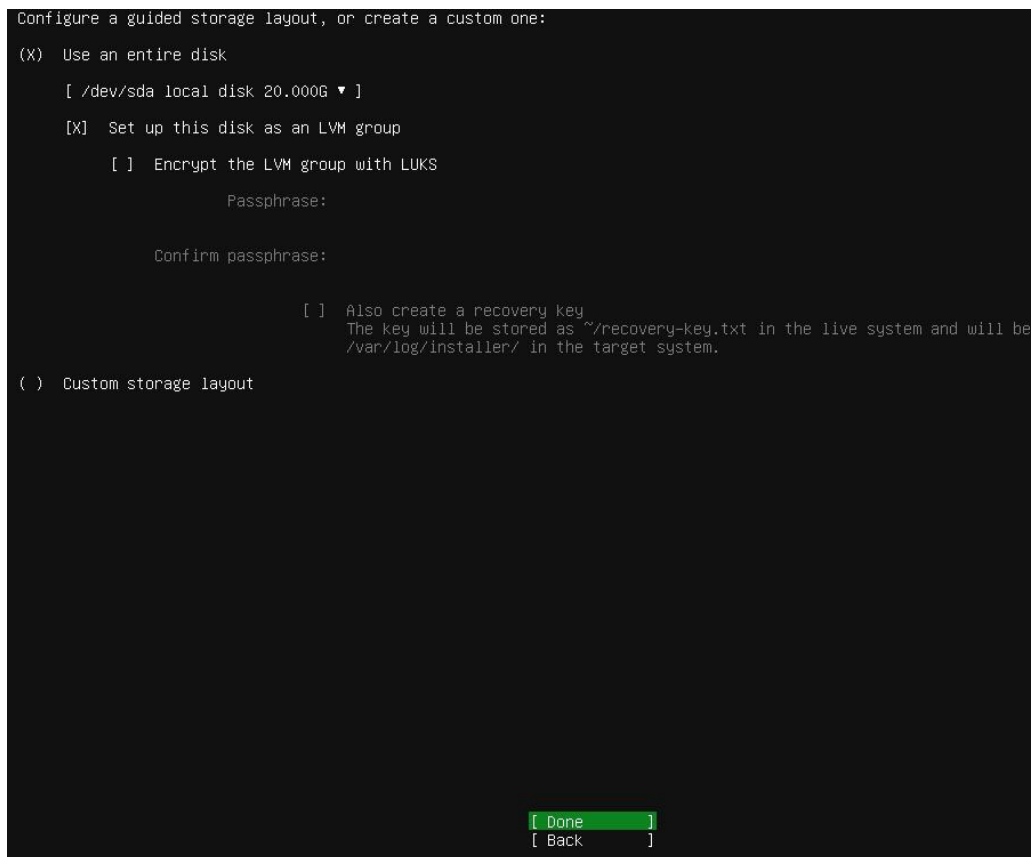


Figure 10.1: Selected Installation Disk (ubuntu_selected_disk.png)



Figure 10.2: Disk Name and Size Details (ubuntu_disk_details.png)

Task 11

Review the Partitions

Description:

Reviewed the proposed file system summary and partition layout configured by the installer to confirm correct mount point allocations on the VMware virtual drive.

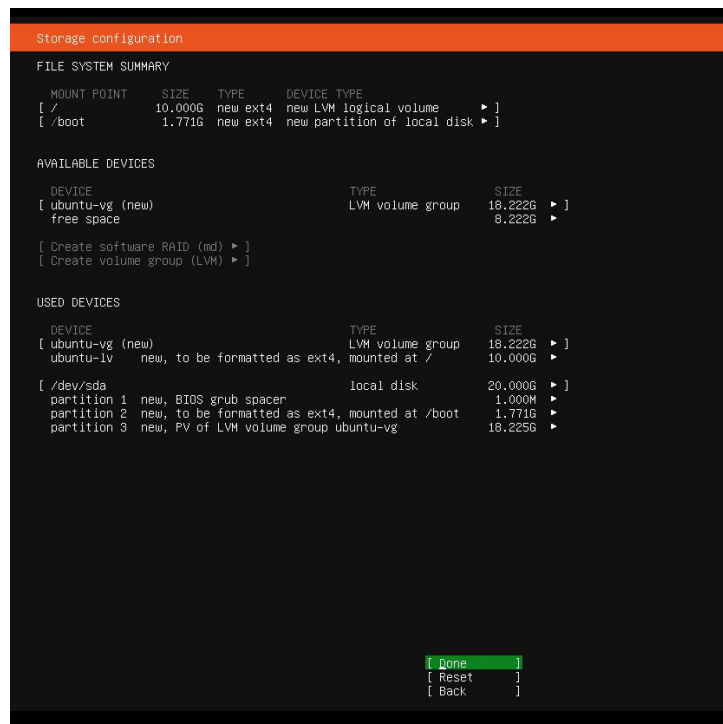


Figure 11.1: Proposed Partition Layout (ubuntu_partition_layout.png)

Task 12

Confirm the Storage Changes

Description:

Acknowledged the destructive action warning dialog regarding disk formatting and confirmed changes to proceed with writing partitions to the virtual disk.

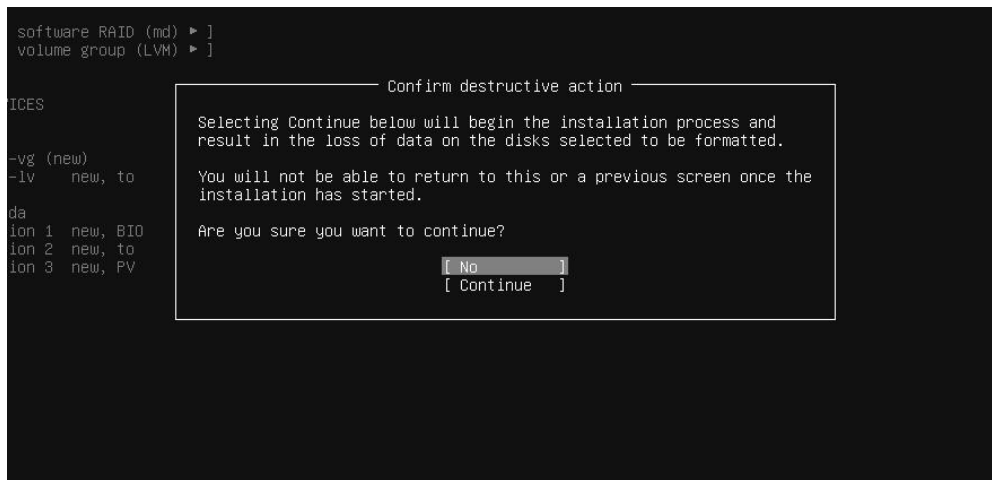


Figure 12.1: Storage Destructive Action Warning (ubuntu_storage_warning.png)

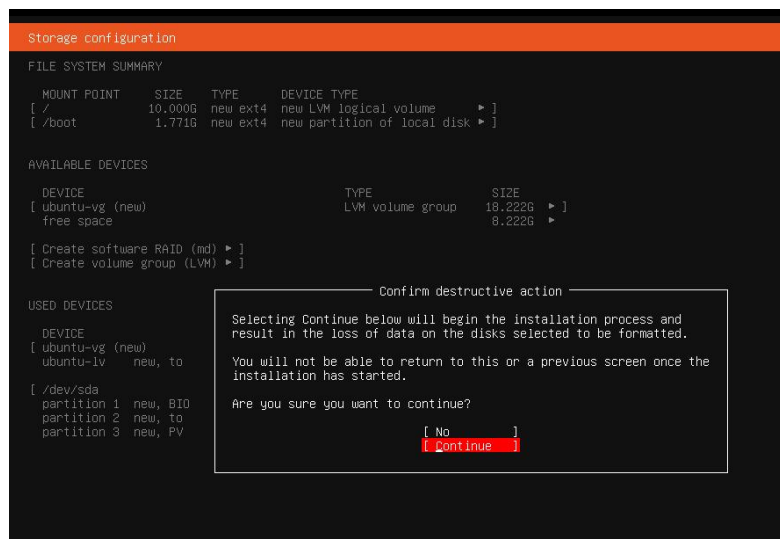


Figure 12.2: Storage Changes Confirmed (ubuntu_storage_confirmed.png)

Task 13

Configure the Ubuntu Profile

Description:

Configured the primary system administrator identity and server hostname settings in the installer using real student details and the designated GitHub handle.



Figure 13.1: Entered Name Field (ubuntu_actual_name.png)



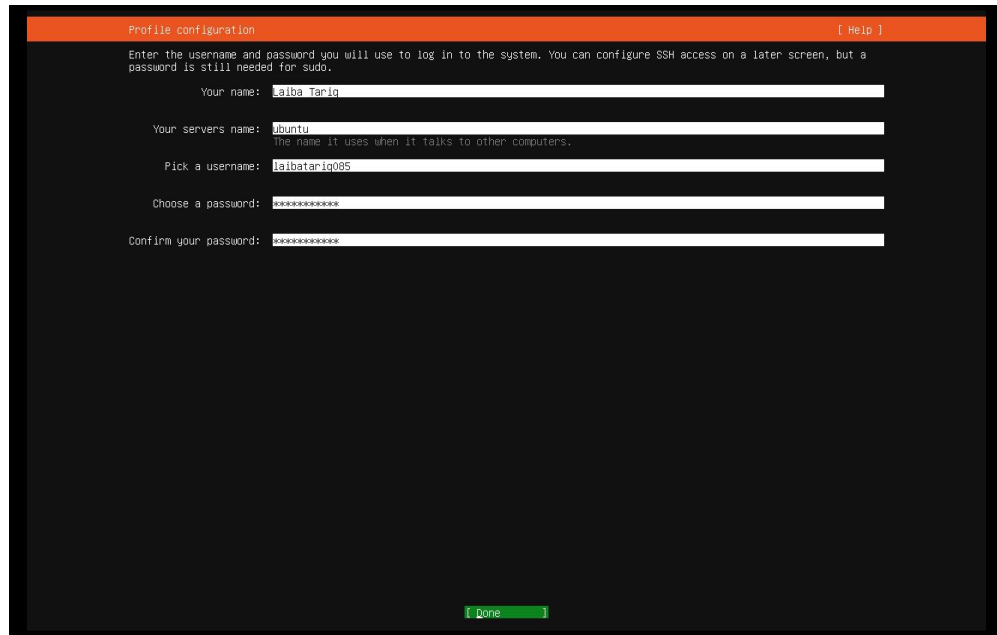
A terminal window showing the configuration of the server's hostname. The prompt is "Your servers name:" followed by the input "ubuntu". Below the input, a message states: "The name it uses when it talks to other computers."

Figure 13.2: Server Hostname Setup (ubuntu_server_name.png)



A terminal window showing the selection of a username. The prompt is "Pick a username:" followed by the input "laibatarig085".

Figure 13.3: Username Configuration (ubuntu_username.png)



A screenshot of the Ubuntu installer's "Profile configuration" screen. The screen has an orange header with the title "Profile configuration" and a "[Help]" link. Below the header, a message reads: "Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo." The form contains several input fields: "Your name:" with the value "Laiba Tariq", "Your servers name:" with the value "ubuntu" and a subtext "The name it uses when it talks to other computers.", "Pick a username:" with the value "laibatarig085", "Choose a password:" with masked characters "*****", and "Confirm your password:" with masked characters "*****". At the bottom right, there is a green button labeled "[Done]".

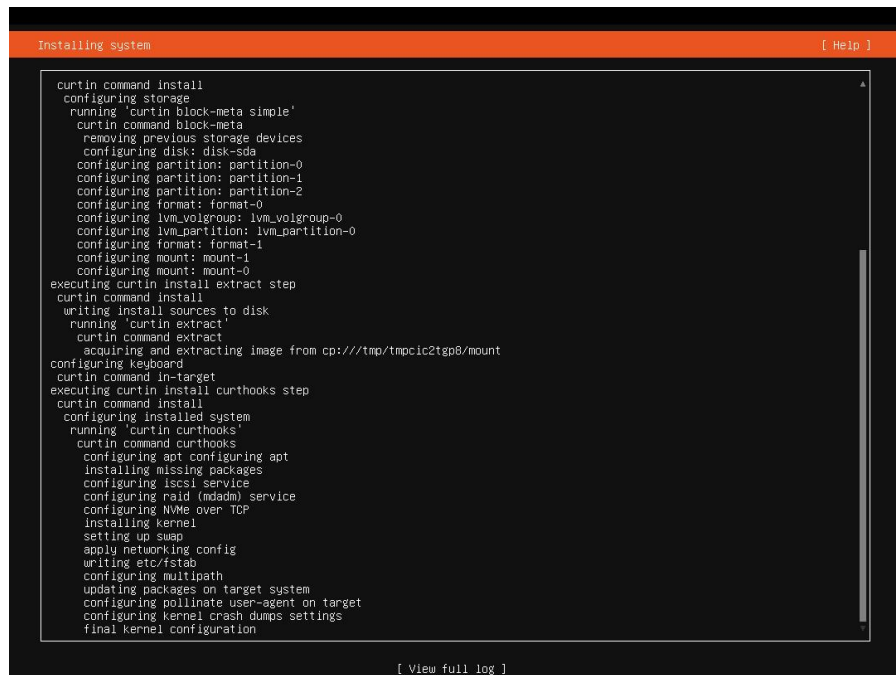
Figure 13.4: Complete Profile Setup Summary (ubuntu_profile_setup.png)

Task 14

Complete the Software Installation

Description:

Initiated the core system software extraction and package deployment process, ensuring uninterrupted execution until base OS installation completed.



```
Installing system [ Help ]

curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring partition: partition-2
  configuring format: format-0
  configuring lvm_volgroup: lvm_volgroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpcic2tgp8/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
```

Figure 14.1: System Installation Progress (ubuntu_installation_progress.png)

Task 15

Reboot and Verify the Ubuntu Identity

Description:

Rebooted the virtual machine post-installation and authenticated via the command-line interface to confirm the custom system hostname and user prompt alignment.

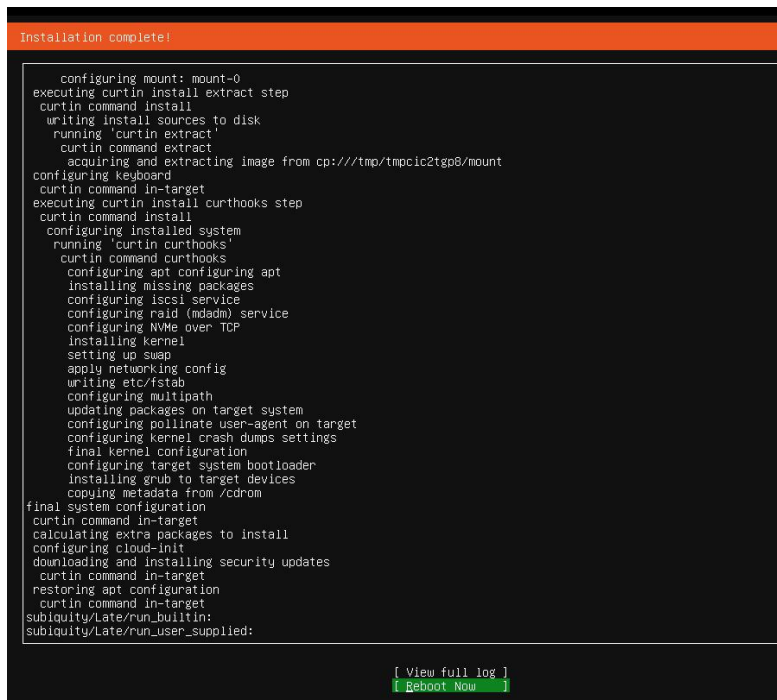


Figure 15.1: Installation Complete Prompt (ubuntu_installation_complete.png)

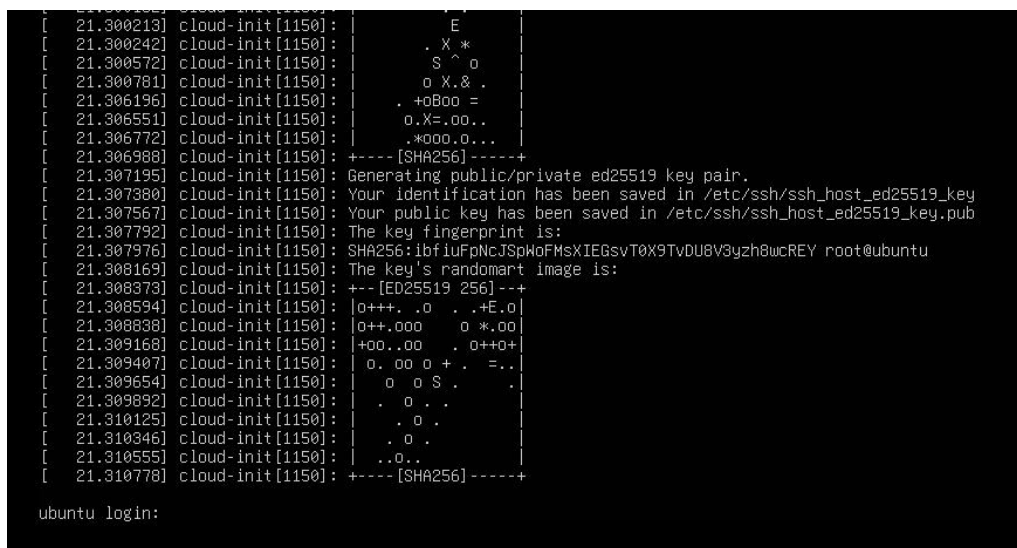


Figure 15.2: Ubuntu TTY Login Screen (ubuntu_login_screen.png)

```

ubuntu login: laibatarig085
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 04:47:42 PM UTC 2026

System load:  0.01          Processes:           249
Usage of /:   50.6% of 9.75GB Users logged in:       0
Memory usage: 22%          IPv4 address for ens33: 192.168.202.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```

Figure 15.3: Successful Terminal Login (ubuntu_terminal_login.png)

```

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibatarig085@ubuntu:~$ _

```

Figure 15.4: Verified User Prompt (ubuntu_identity_verified.png)

Task 16

Find the Ubuntu Server IP Address

Description:

Executed network interface diagnostic commands within the terminal to identify the dynamically assigned IPv4 address on the active virtual interface.

```

laibatarig085@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:55:6a:80 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29556a80
    inet 192.168.202.129/24 metric 100 brd 192.168.202.255 scope global dynamic ens33
        valid_lft 1686sec preferred_lft 1686sec
    inet6 fe80::20c:29ff:fe55:6a80/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Figure 16.1: IP Address Command Output (ubuntu_ip_addr_command.png)

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:55:6a:80 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx000c29556a80
    inet 192.168.202.129/24 metric 100 brd 192.168.202.255 scope global dynamic ens33
        valid_lft 1686sec preferred_lft 1686sec
    inet6 fe80::20c:29ff:fe55:6a80/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
laibatarig085@ubuntu:~$ _
```

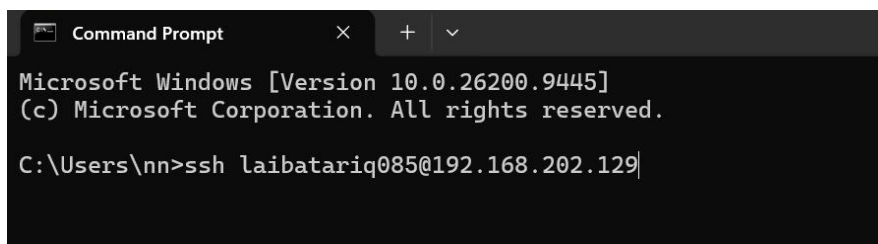
Figure 16.2: Assigned IPv4 Address (ubuntu_ip_address.png)

Task 17

Connect to Ubuntu Server from Windows

Description:

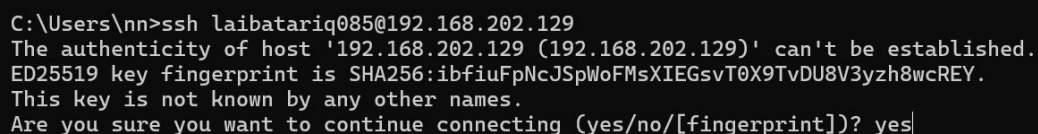
Established a remote SSH CLI session from the Windows host system using PowerShell/Command Prompt to connect to the Ubuntu VM IP address, verified ECDSA host key fingerprint, and authenticated securely.



```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nn>ssh laibatarig085@192.168.202.129
```

Figure 17.1: SSH Connection Command (windows_ssh_command.png)



```
C:\Users\nn>ssh laibatarig085@192.168.202.129
The authenticity of host '192.168.202.129 (192.168.202.129)' can't be established.
ED25519 key fingerprint is SHA256:ibfiuFpNcJSpWoFMsXIEGsvT0X9TvDU8V3yzh8wcREY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Figure 17.2: Accepted Server Fingerprint (windows_ssh_fingerprint.png)

```
laibatariq085@192.168.202.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 05:19:26 PM UTC 2026

System load:  0.0           Processes:            224
Usage of /:   51.1% of 9.75GB Users logged in:      0
Memory usage: 26%          IPv4 address for ens33: 192.168.202.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Figure 17.3: Successful Remote SSH Session (windows_ssh_login.png)

```
C:\Users\mn>ssh laibatariq085@192.168.202.129
laibatariq085@192.168.202.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 05:19:26 PM UTC 2026

System load:  0.0           Processes:            224
Usage of /:   51.1% of 9.75GB Users logged in:      0
Memory usage: 26%          IPv4 address for ens33: 192.168.202.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

laibatariq085@ubuntu:~$ |
```

Figure 17.4: Verified Remote Shell Prompt (windows_ssh_identity.png)

Task 18

Prepare Lab1 Solution PDF

Description:

Compiled all previous task descriptions, structural captions, and verified screenshots into a single, professionally styled document with an official Title Page, exported the final report to PDF format, and structured local submission directories for automated evaluation.

❖ **Lab Number & Title:** Lab 01 - Ubuntu Server Installation

- ❖ **Student Name:** Laiba Tariq
- ❖ **Registration Number:** 2024-BSE-85
- ❖ **Course Code:** SE-665
- ❖ **Section:** Section B
- ❖ **GitHub Username:** LaibaTariq085



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Figure 18.1: Document Title Page (solution_title_page.png)